Ions and Compound Formation

Atoms consist of a central ***nucleus*** made up of positively charged ***protons*** and ***neutrons*** which have no charge. The nucleus is orbited at high speed by much smaller negatively charged ***electrons***. Protons and neutrons have the same mass, while electrons have about 1/2000th the mass of protons and neutrons.

An atom has an equal number of protons and electrons, so the positive and negative charges balance each other to produce a neutral atom (one with no overall charge). The simplest atom is hydrogen, whose nucleus contains only a single proton, orbited by a lone electron. All other atoms contain the three types of particles.

Scientists use two special numbers to describe atoms. The ***atomic number*** of an atom is the number of protons it contains and the ***mass number*** is the number of particles in an atom's nucleus (the number of protons + neutrons). The atomic numbers of hydrogen, helium, lithium and beryllium are 1, 2, 3 and 4 respectively. Scientists use the following notation to describe an atom's structure:

If we know the atomic and mass numbers of an atom, we can figure out how many protons, electrons and neutrons it contains. For example, potassium

has an atomic number of 19 and a mass number of 39. This means there are twenty protons in the nucleus. Because the number of electrons is the same as the number of protons, there must be 19 electrons whizzing around the outside of the atom. The mass number of 29 tells us there are 29 particles in the nucleus, but since we know 19 of these are protons, the other ten must be neutrons.

Electron Shells
Electrons orbit the nucleus in special regions, or shells. Only 2 electrons may fit in the innermost shell of any atom, then up to 8 in each of the next 2 shells for elements up to atomic number 20. The inner most shells are 'filled' first. This may be compare to filling spaces on a bookshelf, where we fill the bottom shelf first, then only move up if each lower shelf is filled.

Take sodium, for example, with 11 electrons. 2 electrons orbit in the inner shell, 8 in the next, leaving one to orbit in the third shell.

Atoms can become charged if they gain or lose electrons. Charged atoms are called ***ions***. For example, sodium has only one electron in its outermost shell. This means that sodium will readily give up this electron. When it loses this electron, sodium becomes positively charged. Conversely, the chlorine atom forms a negative ion when it gains an electron.

When sodium and chloride are able to exchange an electron, an ionic bond forms between them and the compound, sodium chloride, is formed. The ***molecular formula*** for this is: NaCl. Magnesium readily gives up its 2 outermost electrons, forming Mg^{2+}. So when magnesium combines with chloride the molecular formula is $MgCl_2$, because 2 chloride ions are needed for each magnesium ion in order to balance the charge.

Questions

1. Write a definition for each of the following:

Nucleus:

_____
_____

Proton:

_____
_____

Neutron

_____
_____

Electron

_____
_____

Atomic number

_____
_____

Mass number

_____
_____

Ion

_____
_____

Molecular formula

_____
_____

2. Give the symbols for the following elements:

(a)	Carbon	(c)	Gold
(b)	Aluminium	(d)	Tin

3. Give the names of the elements that have these symbols:

(a)	Ag	(c)	Cu
(b)	Fe	(d)	B

4. How many protons, neutrons and electrons are there in each of the following atoms?

(a) Fe
(b) Cu
(c) I
(d) U

5. Complete the table below:

Atom	Gains or loses electrons	Ion
Fluoride	Gains 1 electron	F^-
Zinc	Loses 2 electrons	
Oxygen	Gains 2 electrons	
Sodium	Loses 1 electron	
Silver	Loses 1 electron	
Chlorine	Gains 1 electron	

6. Complete the following using the word list below:

E_e_e_ _ A pure substance.

_to_ The smallest part of an element.

Mol_ _ _ _ _ Group of atoms bonded together.

_ _ _ pound Substance made up of identical molecules.

Form_ _ _ A short way of describing the elements in a compound.

_ _ x _ _ _ _ Combination of elements and /or compounds not bonded together.

_ _ c _ _ _ _ The centre of an atom.

_ rot _ _ Positively charged parts of an atom.

N _ _ _ _ _ _ Have no charge.

_ _ _ _ _ _ _ _ Negatively charged parts of an atom.

_ _ n An atom that has lost or gained electrons.

Word List
ion – mixture – compound – formula – proton – nucleus – electron
element – neutron – molecule – atom.